

JEE MAIN REVISION MASTER: APPLICATION OF DERIVATIVES SYLLABUS REDUCED

Subject: Mathematics | Target Scope: JEE Main Previous Years | Empirical Base: 227 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
Very High	Application Of Derivatives - Core Application	227 PYQs	General Concept

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
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3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 Application Of Derivatives - Core Application (Very High Yield — 227 PYQs)

Core Principle: Key problem pattern in Application Of Derivatives.

■ ■ Caution / JEE Pitfall: Watch for algebraic and sign errors.

■ Speed Trick: Verify dimensions and boundary conditions.

Representative Source PYQs:

- [2026 Morning Shift] Q#1 (MCQ): "Let $f: \mathbb{R} \rightarrow \mathbb{R}$ be a differentiable function such that $f((x+y)/3) = (f(x)+f(y))/3$ for all $x, y \in \mathbb{R}$, and $f'(0)=3$..." [Ans: B]
- [2026 Evening Shift] Q#1 (MCQ): "The least value of $(\cos^2\theta - 6\sin\theta \cos\theta + 3\sin^2\theta + 2)$ is..." [Ans: N/A]
- [2026 Morning Shift] Q#1 (Numerical): "Let $(2\alpha, \alpha)$ be the largest interval in which the function $f(t) = \frac{|t+1|}{t^2}$, $t < 0$, is strictly decreasing. Then the local..." [Ans: 4]

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: JEE Main 2026 (Online) 5th April Morning Shift (Q#1)

Let $f: \mathbb{R} \rightarrow \mathbb{R}$ be a differentiable function such that $f\left(\frac{x+y}{3}\right) = \frac{f(x)+f(y)}{3}$ for all $x, y \in \mathbb{R}$, and $f'(0)=3$. Then the minimum value of the function $g(x)=3+e^x f(x)$, is:

- (A) $3\left(\frac{e+1}{e}\right)$
- (B) $3\left(\frac{e-1}{e}\right)$ **[Correct]**
- (C) $(3-e)/e$
- (D) $3e$

Correct Answer: B

Solution Analysis: Given $f\left(\frac{x+y}{3}\right) = \frac{f(x)+f(y)}{3}$ for all $x, y \in \mathbb{R}$, and $f'(0)=3$. We have to find the minimum value of $g(x)=3+e^x f(x)$. Step 1: Find the form of $f(x)$. We are given a functional equation. We will use it carefully. Let $\frac{x+y}{3}=t$. Then $x+y=3t$. The relation becomes $f(t) = \frac{f(x)+f(y)}{3}$ whenever $x+y=3t$. We now choose suitable values. Put $x=y=0$. Then $f(0) = \frac{f(0)+f(0)}{3} = \frac{2f(0)}{3}$. So, $3f(0) = 2f(0) \Rightarrow f(0)=0$. Put $y=0$. Then $f\left(\frac{x}{3}\right) = \frac{f(x)+f(0)}{3} = \frac{f(x)}{3}$. Thus, $f(x) = 3f\left(\frac{x}{3}\right)$. Now differentiate both sides with resp